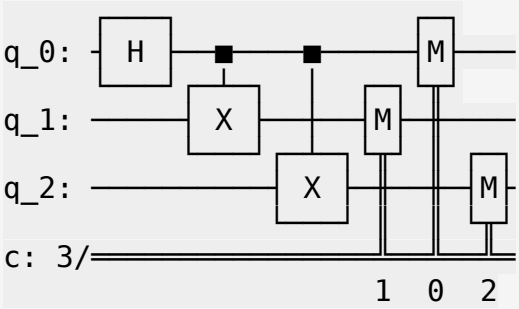
Practical 4

```python
from qiskit import QuantumCircuit, transpile
from qiskit_aer import AerSimulator

qc = QuantumCircuit(3, 3)

# Create superposition
qc.h(0)

# Entanglement
qc.cx(0, 1)
qc.cx(0, 2)

# Measure
qc.measure([0, 1, 2], [0, 1, 2])

simulator = AerSimulator()

compiled = transpile(qc, simulator)

result = simulator.run(compiled, shots=1000).result()

counts = result.get_counts()

print("Measurement Counts:")
print(counts)

print("\nCircuit:")
print(qc.draw())
```

```
Measurement Counts:
{'111': 507, '000': 493}

Circuit:
```